
Masking Cannot Remove What Is Inside the Region of Interest: Location-Dependent Shortcut Mitigation in Medical Imaging

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Abstract

1 Medical image classifiers exploit acquisition artifacts—hair, sonographer calipers,
2 intestinal debris—that correlate with the diagnosis; the standard defence restricts
3 the model to the region of interest (ROI). We show that this defence fails for arti-
4 facts inside the ROI, give a closed-form account of when and why, and evaluate
5 alternatives. In pre-registered, leakage-controlled traps built from real artifacts
6 in four diseases, masking always helps more for out-of-ROI than for in-ROI arti-
7 facts (crossover $+0.15$ to $+0.37$ reversed-test AUROC, all Holm-significant), with
8 frozen and fine-tuned backbones and in the zero-shot scores of vision–language
9 foundation models. Masking becomes harmful when it also removes disease con-
10 text: in dermoscopy, controlled overlap sweeps, fine-tuning (capsule -0.28) and
11 unaltered test data (official thyroid split -0.10 on shortcut-conflicting cases; two
12 of three held-out dermoscopy hospitals). A linear-Gaussian model fitted only to
13 each classifier’s clean and correlated performance matches its held-out reversed-
14 test AUROC (mean absolute error 0.039) and the location crossover across 14
15 cohort–backbone pairs ($r = 0.997$). For frozen foundation-model features we
16 propose mask-then-erase with generic synthetic overlays (U-MtE), which needs
17 no artifact examples, artifact masks or artifact labels—only the ROI mask. It re-
18 moves in-ROI shortcuts carried by distinct overlays (thyroid $+0.13$ to $+0.22$ over
19 masking on three backbones); erasure beats augmentation with the same overlays
20 (thyroid, dermoscopy); a disease-protected variant prevents failure on pathology-
21 like artifacts; text-prompted erasure is a near no-op. Shortcuts carried by cor-
22 relates of the artifact (hair, chest drains; chest radiography is a boundary case)
23 require label-based reweighting. Code, pre-registrations and a backbone-agnostic
24 tool are public.

25 1 Introduction

26 Deep classifiers for medical images rely on features that correlate with the label but are not evi-
27 dence of disease [Geirhos et al., 2020, DeGrave et al., 2021]: surgical markings and rulers change
28 melanoma predictions [Winkler et al., 2019, 2021], and pneumothorax classifiers detect the chest
29 drain that treats the condition [Oakden-Rayner et al., 2020, Jiménez-Sánchez et al., 2023]. The most
30 common defence is to restrict the classifier to the region of interest (ROI) by masking the back-
31 ground. Its rationale assumes that shortcuts live outside the ROI. Hair, calipers and debris do not
32 respect that assumption: when the artifact overlaps the ROI, masking cannot remove it—it removes
33 everything *else*.

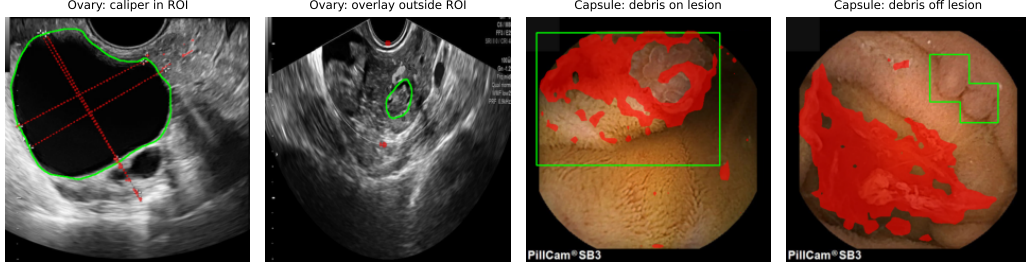


Figure 1: Real artifacts inside and outside the region of interest (green: ROI; red: artifact mask). Masking the background removes an out-of-ROI artifact (2nd, 4th) but keeps an in-ROI one (1st, 3rd) while deleting the context around it. Left: ovarian tumours (MMOTU); right: capsule endoscopy (SEE-AI).

We ask whether the value of a mitigation depends on where the artifact sits, which quantity governs that dependence, and how in-ROI shortcuts can be removed without localising the artifact. Our contributions:

1. **A location law for ROI masking**, established with pre-registered real-artifact traps in four diseases (dermoscopic hair; thyroid and ovarian sonographer calipers; capsule-endoscopy debris), replicated with three frozen backbones, two fine-tuned architectures, the zero-shot scores of vision–language models and on unaltered test data.
2. **A closed-form account** (Sec. 5) of when in-ROI masking harms, when erasure fails, and why group reweighting is location-invariant, which matches held-out results of real encoders in a pre-registered test.
3. **U-MtE**, a representation-level remedy for frozen foundation-model features that needs no artifact annotation (only the ROI mask), with evidence that erasure (not augmentation) is what works, a disease-protected variant, and a delineation of the regimes in which it does not apply.
4. **Rigour**: pre-registration before every experiment, Holm-corrected primary claims (12/16 supported), replication of an earlier pilot (73/88 numbers), and gaming-aware reporting.

2 Related work

Bissoto et al. [2020] showed that artifact debiasing in dermoscopy is harder than it appears and built trap sets; test-time selection [Bissoto et al., 2023] and generative repair [Jin et al., 2024] address specific artifacts. Hair occlusion degrades HAM10000 classifiers [Wang et al., 2024]; adding synthetic dark-corner artifacts to training improves classification [Pewton et al., 2024]; calipers are shortcuts even for fetal-ultrasound segmentation [Lin et al., 2024]; and classifiers on frozen foundation-model features show shortcut-like behaviour that pooled evaluation hides [Germani et al., 2026]. None treats the artifact’s location relative to the ROI as a variable. Group-robust training [Sagawa et al., 2020, Kirichenko et al., 2023, Liu et al., 2021] needs group labels or error sets; linear concept erasure [Ravfogel et al., 2020, Belrose et al., 2023] and its task-preserving form SPLINCE [Holstege et al., 2025] need concept labels; biased-prompt projection [Chuang et al., 2023] derives directions from text. Random erasing and occlusion augmentation [Zhong et al., 2020, Fong and Vedaldi, 2019] use overlays as training noise. We estimate an artifact subspace from synthetic *insertion* pairs and remove it only after masking.

3 Setup

Traps Let Y be the diagnosis and A the presence of an artifact. A trap trains with $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$ and tests on a correlated (same rates), a reversed (swapped) and a clean environment (artifact independent of the label: present in half of each class in real traps, absent in sweeps). The overlap $r = |\text{artifact} \cap \text{ROI}|/|\text{artifact}|$ defines Trap A (in-ROI, $r \geq 0.5$) and

Table 1: Real-artifact cohorts. Artifact masks: published (Wegley et al.), audited detector (95% precision on 40 images), or expert-trained probe (pixel accuracy 0.92).

Cohort	Modality	Images	Artifact (mask source)	ROI
ISIC 2019	dermoscopy	25,331	hair/ruler (published masks)	lesion
TN3K/TNCD	thyroid ultrasound	3,493	calipers (rule-based detector)	nodule
MMOTU	ovarian ultrasound	1,202	calipers/overlays (same detector)	tumour
SEE-AI	capsule endoscopy	5,481	debris/bubbles (patch-token probe)	lesion boxes
NIH ChestX-ray14	radiography	29,687	chest drains (image-level)	lungs

Trap B (out-of-ROI, $r < 0.1$); both share one artifact-free group, cells are matched on source and label, and splits are grouped by lesion/patient identifiers where published, otherwise by perceptual hashes and frame blocks (five seeds $\times$ five folds; a stricter embedding-based grouping is a sensitivity analysis). The primary metric is reversed-test AUROC; we always report correlated and clean AUROC, and summarise robustness by $\min(\text{reversed, correlated})$, which penalises methods that merely flip the shortcut. Uncertainty is a hierarchical paired bootstrap (10,000 replicates, clusters = seeds or folds).

Cohorts (Table 1; details in Appendix A.) Heads are linear on frozen DINOv2 ViT-B/14, MedSigLIP, DermLIP, RAD-DINO or ConvNeXt features; regularisation and threshold are chosen on clean validation data only. Controlled sweeps paste a synthetic artifact (ruler, caliper, debris, tube) at overlap $r \in \{0, 0.25, 0.5, 0.75, 1\}$.

4 Mask-then-erase with a generic insertion library

Removing an in-ROI artifact requires its pixels; inserting one does not. A procedural generator $\mathcal{G}$ (strokes, bands, patches, blobs, dashed lines, glyphs; random colour, opacity and position, half centred in the ROI) turns every masked training image $m(x_i)$ into a counterfactual $m(\mathcal{G}(x_i))$; the feature differences $\Delta_i = f(m(\mathcal{G}(x_i))) - f(m(x_i))$ are independent of Y . U-MtE erases their span,

$$U_k = \text{top-}k \text{ eigvecs} \left(\frac{1}{n} \sum_i \Delta_i \Delta_i^\top \right), \quad P(z) = z - U_k U_k^\top (z - \mu), \quad (1)$$

with k chosen without labels (90% held-out difference energy), and fits the head on $P(f(m(x)))$. The generator contains no ruler, hair, caliper or debris template. *Disease protection* erases $U'_k = \text{orth}((I - WW^\top)U_k)$, where W spans logistic label directions fitted on artifact-free training images, guaranteeing $w^\top P(z) = w^\top z$ for $w \in \text{span}(W)$. With image-level artifact labels the head is group-balanced (U-MtE_{bal}).

5 Why location matters

Let the disease block have class-separation SNR S and the artifact block SNR A per unit of presence. For the LDA-optimal linear head trained with (p_1, p_0) , the artifact’s effective SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}} = \Phi \left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}} \right), \quad (2)$$

increasing in S and decreasing in $\tilde{A}$ (simulation agreement ≤ 0.009 AUROC; Appendix B). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$ with $\tilde{A}_m = 0$ out of the ROI and $\tilde{A}_m \geq \tilde{A}$ inside. Hence in-ROI masking harms exactly when it removes disease context ($S_m < S$); the crossover is positive for every S_m ; erasure sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$ (ρ : overlap of the erased subspace with the disease direction), so it fails for pathology-like artifacts unless protected; and group reweighting yields $\Phi(\sqrt{S/2})$ regardless of location (simulation in Appendix Fig. 5).

Test on real encoders The same head gives the AUROC of any test composition (Appendix B). In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC only,

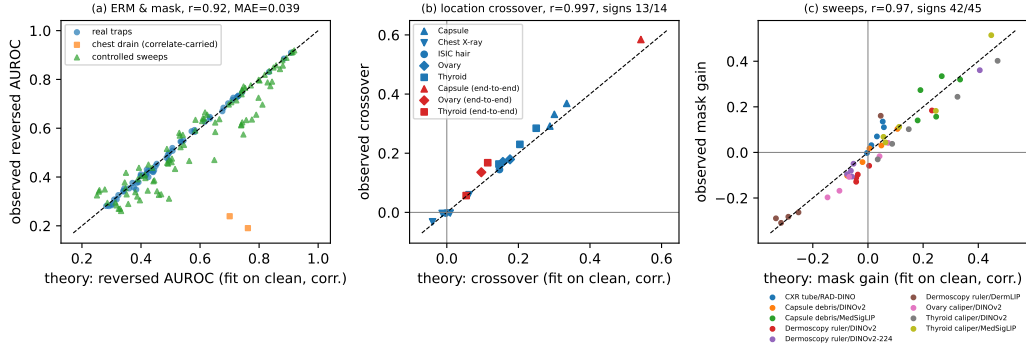


Figure 2: Pre-registered test of the closed form on real encoders: (S, A) fitted to each model’s clean and correlated AUROC, reversed-test results held out. (a) Reversed-test AUROC (ERM, masking). (b) Location crossover per cohort and backbone (red: end-to-end fine-tuned, outside the model’s assumptions). (c) Mask gain along controlled sweeps.

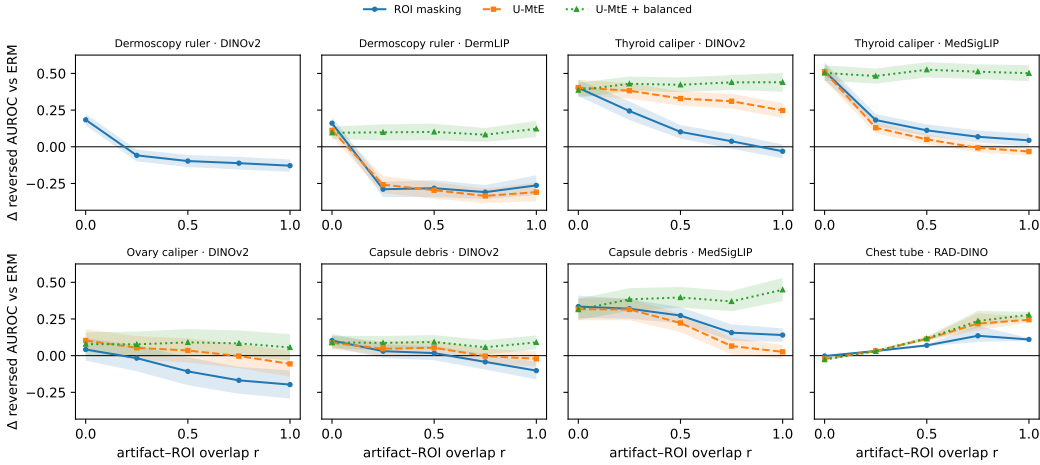


Figure 3: Controlled overlap sweeps: Δ reversed-test AUROC versus ERM as a synthetic artifact moves into the ROI (95% CIs). Masking (blue) declines with overlap except in chest radiography, where the encoder ignores extra-pulmonary tubes; U-MtE with group balancing (green) is flat.

with no shared or tuned parameter, and compared the implied reversed-test AUROC with the held-out observation (Fig. 2): mean absolute error 0.039 over 148 ERM and masking models ($r = 0.92$; 0.108 for a linear-symmetry heuristic). The implied location crossover agrees with the observed one across 14 cohort–backbone pairs ($r = 0.997$, error 0.014, 13/14 signs; the miss is a chest-radiograph cell at -0.002) and also for fine-tuned networks (4/4 signs). One sign miss means that, by our pre-registered criterion, the model remains a qualitative account; it fails only for chest drains, whose shortcut is carried by correlates that the clean and correlated environments do not expose.

6 Results

6.1 A location law for ROI masking

In all four real-artifact cohorts the crossover is positive and survives Holm correction (Table 2); it replicates with MedSigLIP and ConvNeXt, with end-to-end fine-tuning (ResNet-50 capsule; ViT-S thyroid), and in the zero-shot scores of MedSigLIP and DermLIP, i.e. before any task training. Masking becomes *harmful* for in-ROI artifacts when it also removes disease context: real dermoscopic hair (-0.037 [$-0.057, -0.019$]), controlled sweeps at full overlap (ovary -0.197 [$-0.292, -0.100$];

Table 2: Masking gain (mask – ERM, reversed AUROC) is always larger for out-of-ROI than in-ROI artifacts. Crossover = gain(out) – gain(in), 95% CI. Zero-shot: no task training. Natural: shortcut-conflicting pairs of untouched test data.

Setting	Cohort (backbone)	Crossover [95% CI]
Real traps, frozen	ISIC hair (DINOv2)	+0.145 [+0.105, +0.181]
	Thyroid (DINOv2 / MedSigLIP / ConvNeXt)	+0.230 / +0.163 / +0.284
	Ovary (DINOv2)	+0.170 [+0.116, +0.224]
	Capsule (DINOv2 / MedSigLIP / ConvNeXt)	+0.368 / +0.332 / +0.292
Fine-tuned	Capsule (ResNet-50)	+0.584 [+0.469, +0.691]
	Thyroid (ViT-S)	+0.168 [+0.074, +0.265]
Zero-shot VLM	Thyroid / Capsule (MedSigLIP)	+0.103 / +0.049
	Ovary (MedSigLIP) / Hair (DermLIP)	+0.181 / +0.051
Natural, mask – ERM	Thyroid (official split)	–0.102 [–0.139, –0.066]
	ISIC held-out BCN / MSK	–0.016 / –0.056

Table 3: Unaltered data (no resampling): change in AUROC on shortcut-conflicting pairs. Thyroid: official TN3K split; ISIC: each hospital held out in turn; capsule: held-out frame blocks. 95% CIs.

Test set	mask – ERM	U-MtE – mask	U-MtE _{bal} – mask
Thyroid (official split)	–0.102 [–0.139, –0.066]	+0.051 [+0.033, +0.072]	+0.088 [+0.071, +0.108]
ISIC, BCN held out	–0.016 [–0.027, –0.005]	+0.003 [+0.000, +0.006]	+0.034 [+0.029, +0.039]
ISIC, MSK held out	–0.056 [–0.084, –0.029]	–0.008 [–0.015, –0.002]	+0.041 [+0.029, +0.053]
ISIC, HAM held out	+0.053 [+0.037, +0.068]	+0.023 [+0.017, +0.032]	+0.065 [+0.056, +0.075]
Capsule (held-out frames)	+0.021 [+0.000, +0.044]	+0.000 [–0.010, +0.009]	+0.014 [+0.003, +0.025]

capsule –0.102 [–0.162, –0.040]; Fig. 3), fine-tuned capsule (–0.284 [–0.347, –0.226]) and on unaltered data, where masking lowers AUROC on shortcut-conflicting cases for thyroid and two of three held-out ISIC hospitals (Table 3); at the validation-fixed operating point it halves thyroid sensitivity (0.734 → 0.347; AUPRC, Brier score and calibration in Appendix Table 6). The claims are robust to the artifact-label definition (large-marker and strict-location variants) to a visual audit of the masks and to a stricter, embedding-based leakage grouping (11 of 12 primary estimates unchanged; Appendix E). The law also holds across model scale: with DINOv2 ViT-S, ViT-B and ViT-L the crossover is positive in all nine cohort-size pairs (+0.17 to +0.41), and larger models exploit in-ROI calipers *more* (ERM correlated–reversed gap 0.547 → 0.621 on thyroid, 0.319 → 0.570 on ovary from ViT-S to ViT-L). Chest radiography is a boundary case: the encoder ignores extra-pulmonary tubes, so masking has nothing to remove outside the lungs.

Mechanism. Three explanations were tested separately in the controlled dermoscopy sweep (Appendix C). *Occlusion* is not the cause: with the artifact present but uncorrelated with the label, masking helps at every overlap. *Lost context* alone is not the cause: dilating the mask at 50% overlap increases harm as it restores artifact pixels, and at full overlap harm is flat although the artifact’s share of the visible pixels falls threefold. The governing quantity is artifact *retention*: masking keeps an in-ROI artifact, and the masked model’s counterfactual sensitivity to it roughly doubles.

6.2 Removing in-ROI shortcuts

Without any artifact example, artifact mask or label, U-MtE improves on masking where the artifact is a distinct overlay: thyroid +0.222 [+0.197, +0.251] (DINOv2), +0.204 [+0.174, +0.237] (MedSigLIP), +0.128 [+0.112, +0.145] (ConvNeXt); ovary +0.215 [+0.182, +0.246] (MedSigLIP); controlled sweeps +0.08 to +0.28; end-to-end (ResNet-50, thyroid) +0.143 [+0.099, +0.188]. The same overlays used as training augmentation gain at most +0.03; erasing their subspace gains +0.207 [+0.182, +0.235] more on thyroid. On capsule debris, which resembles erosion fibrin, unprotected erasure removes disease signal; disease protection removes the failure. With image-level artifact labels, U-MtE_{bal} is the most robust model on thyroid with all three backbones (Table 4; full values in Table 7). The gain is stable across model sizes (protected U-MtE on thyroid: +0.181, +0.217, +0.222 for ViT-S/B/L) and erasure ranks ≥ 16 (+0.18 to +0.22). Applied post hoc to

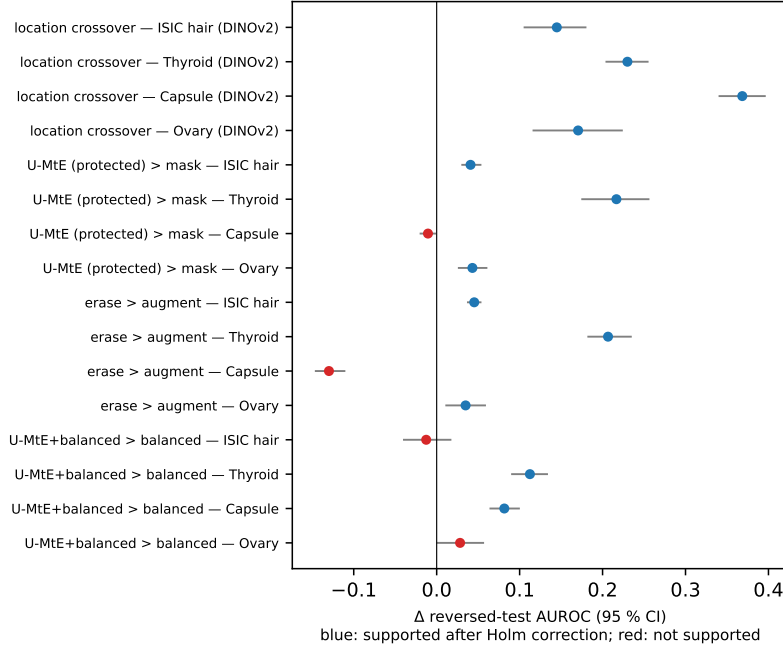


Figure 4: The 16 pre-registered primary tests with Holm correction (blue: supported, 12/16; red: not supported).

Table 4: In-ROI artifacts (Trap A): robustness = $\min(\text{reversed}, \text{correlated})$ AUROC, which penalises both shortcut use and shortcut flipping. Bold: best per column. Full reversed/correlated/clean values for all cohorts and backbones: Table 7.

Method	Needs	Hair DINOv2	Thyroid DINOv2	Thyroid MedSigLIP	Ovary DINOv2	Capsule DINOv2	Drain RAD-DINO
ROI masking	ROI	0.45	0.40	0.35	0.55	0.68	0.24
Mask + overlay augmentation	ROI	0.46	0.42	0.38	0.53	0.70	0.24
U-MtE (ours)	ROI	0.51	0.62	0.55	0.56	0.57	0.27
U-MtE, protected (ours)	ROI	0.49	0.62	0.61	0.59	0.67	0.27
Group-balanced	$\mathcal{A}$	0.72	0.68	0.70	0.71	0.78	0.54
DFR	$\mathcal{A}$	0.74	0.66	0.72	0.69	0.80	0.68
Mask + DFR	ROI + $\mathcal{A}$	0.69	0.74	0.76	0.75	0.88	0.65
U-MtE + balanced (ours)	ROI + $\mathcal{A}$	0.70	0.79	0.79	0.74	0.86	0.46

144 a fine-tuned ResNet-50 it does not help (+0.004), but estimated on the network’s own features
145 and removed *during* fine-tuning, together with overlay consistency, it improves on masking for
146 both ResNet-50 (+0.157 [+0.114, +0.205]) and ViT-S (+0.143 [+0.103, +0.187]) on thyroid (Ap-
147 pendix E.2). Where the shortcut is carried by correlates of the artifact (hair co-varies with body site,
148 age and sex; a drain comes with a treated lung) no pixel- or insertion-based method helps much and
149 DFR [Kirichenko et al., 2023] is best.

150 **Baselines** SPLINCE [Holstege et al., 2025], fitted with artifact labels, removes much disease sig-
151 nal when artifact and label are strongly associated (clean AUROC 0.56–0.78) and flips the short-
152 cut on capsule; U-MtE_{bal} is more robust on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs
153 0.683; 0.704 vs 0.564). Text-prompted erasure (biased prompts, Chuang et al., 2023) is a near no-op
154 ($\leq +0.008$ over masking): the direction along which a *described* artifact moves the text embed-
155 ding is not the one along which the real artifact moves the image embedding. Oracle-mask LaMa
156 inpainting [Suvorov et al., 2022] is strong when artifact masks exist; protected U-MtE matches it
157 on thyroid without masks. Label-free, U-MtE beats JTT [Liu et al., 2021] on thyroid (+0.198)

158 and ovary (+0.137). A validation-driven selector beats masking in all five cohorts but not DFR
159 everywhere (Appendix F).

160 7 Discussion and limitations

161 **What carries the shortcut decides the method** Across six real and controlled settings three
162 regimes emerge. (1) *Distinct overlays that carry the shortcut in their pixels* (sonographer calipers,
163 synthetic rulers, debris and tubes): masking removes an out-of-ROI copy but leaves an in-ROI copy
164 intact, and U-MtE removes most of what masking leaves (thyroid +0.22, +0.20, +0.13 on three
165 backbones; controlled sweeps +0.08 to +0.28). (2) *Artifacts that resemble the pathology* (capsule
166 debris versus erosion fibrin): any subspace estimated from artifact appearance removes disease ev-
167 idence; disease protection makes erasure safe, and with image-level artifact labels masking plus
168 last-layer retraining is the most robust option. (3) *Artifacts whose shortcut is carried by correlates*
169 (dermoscopic hair co-varies with body site, age and sex; a chest drain comes with a treated, re-
170 expanded lung): removing even every artifact patch with an oracle mask gains little (+0.015 for
171 hair), no pixel- or insertion-based method can close the gap, and label-based group methods (DFR,
172 balancing, JTT on hair) are preferable. The oracle-removal ceiling is a cheap diagnostic of the
173 regime.

174 **Why erasure works where masking and patch removal do not** In thyroid ultrasound, dropping
175 every patch token that contains a caliper (oracle masks) before mean-pooling improves the reversed
176 AUROC by only +0.047 over pooling all ROI tokens, whereas erasing the generic-overlay sub-
177 space from the class-token representation improves on masking by +0.22 (the two representations
178 differ, so the comparison is indicative). Self-attention distributes the information about a salient
179 overlay across the whole representation; an in-ROI artifact is therefore not contained in “its” pixels
180 or patches, and a representation-level operation is needed to remove it.

181 **Erasure, not augmentation** The same overlays used as training augmentation gain at most +0.03
182 over masking in every cohort; used to estimate and erase a subspace they gain up to +0.21 more.
183 Augmentation teaches invariance to the *generated* overlays; erasure removes the directions along
184 which *any* overlay moves the representation, which transfers to real artifacts the generator never
185 produced.

186 **Reading reversed-environment gains** A reversed-test gain can be manufactured by removing
187 diagnosis-correlated signal together with the artifact: erasure fitted on unpaired image-level labels
188 and post-hoc prevalence recalibration invert the correlated/reversed ordering. We report reversed,
189 correlated and clean AUROC for every method and summarise robustness by the minimum of corre-
190 lated and reversed AUROC.

191 **Decision rule for practitioners** (1) Measure where artifacts sit relative to the ROI; out-of-ROI
192 artifacts are removed by masking. (2) For in-ROI artifacts without any artifact annotation, use
193 protected U-MtE; it matches oracle-mask LaMa inpainting where the artifact is a distinct overlay.
194 (3) With image-level artifact labels, combine masking or U-MtE with group balancing or last-layer
195 retraining, choosing among them on a group-aware validation split; prefer label-based reweighting
196 alone when the shortcut is carried by correlates of the artifact, which the oracle-removal ceiling (a
197 handful of annotated masks) reveals.

198 **Limitations** (i) U-MtE is a representation-level intervention for frozen (foundation-model) fea-
199 tures with linear heads, the setting in which it is evaluated throughout; end-to-end fine-tuning is a
200 secondary check (the location law replicates; post-hoc erasure of a fine-tuned network’s features
201 does not help, whereas erasing during fine-tuning works for ResNet-50 and ViT-S on thyroid), not
202 a full sweep. (ii) Artifact masks for thyroid calipers and capsule debris come from an audited rule-
203 based detector (precision/recall 0.86–0.89) and a patch-token probe (IoU 0.62); errors act as label
204 noise on A and on the overlap strata. (iii) Hair and ruler share one annotation channel in the ISIC
205 2019 archive. (iv) No patient or video identifiers are published for TN3K, MMOTU or SEE-AI; leak-
206 age groups use perceptual hashes and frame blocks, and a stricter embedding-based grouping leaves
207 11 of 12 primary estimates unchanged (Sec. E.6). (v) Chest-drain labels for pneumothorax-negative

208 radiographs come from a detector. (vi) The controlled chest-X-ray sweep did not show the sign re-
 209 versal: the encoder ignored extra-pulmonary tubes. (vii) Reversed environments are constructed by
 210 subsampling; real deployment shifts are less extreme, and the clean results bound the cost without
 211 shift. (viii) The disease-protection step needs diagnosis labels of artifact-free images, and the bal-
 212 anced variants need image-level artifact labels. (ix) The linear-Gaussian account is fitted per model;
 213 it cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not
 214 expose (chest drains).

215 **Broader impact** Shortcut audits that report only overall accuracy can certify masked models that
 216 still exploit in-ROI artifacts, and a mitigation that flips a shortcut can look like a success on a
 217 reversed test. We recommend reporting artifact location relative to the ROI and performance on
 218 shortcut-conflicting cases together with clean and correlated performance.

219 8 Conclusion

220 ROI masking is not neutral: it removes shortcuts outside the ROI and leaves—or amplifies—those
 221 inside it. This holds across diseases, modalities, training regimes and foundation models, and a two-
 222 parameter model fitted to clean and correlated performance matches the held-out results. Erasing
 223 a subspace learned from generic synthetic overlays removes in-ROI shortcuts carried by distinct
 224 artifacts without any annotation of the artifact.

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A Data

Controlled dermoscopy cohort ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting labels dropped); perceptual-hash (Hamming ≤ 8) and lesion grouping; 2,437 images for which a 65×19 px ruler (0.46% of a 518×518 image) can be placed at 0/25/50/75/100% overlap with the lesion within 10 percentage points (train/val/test 1,849/371/371 before common support).

Real hair cohort (ISIC 2019) All 25,331 ISIC 2019 training images; hair/ruler masks from the archive of Wegley et al. [2026]; lesion masks manual for HAM10000 [Tschandl et al., 2018] and predicted elsewhere by a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels at native resolution): Trap A contains hair with overlap $r = |\text{hair} \cap \text{lesion}|/|\text{hair}| \geq 0.5$, Trap B $r < 0.1$. Cells are subsampled to the hair-free source mix (HAM/BCN/MSK) within each label, giving Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma); five seeds of five-fold group-safe cross-validation (leakage groups: lesion identifiers and pHash); 172 reversed-test melanomas per trap and seed.

Thyroid ultrasound TN3K images with TNCD benign/malignant labels (3,493 images; 1,210 malignant) and manual nodule masks. Sonographer calipers were segmented by a frozen rule-based detector (dotted measurement lines and ‘+’ ends; audit in Sec. 7). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801. No patient identifiers are published; leakage groups are pHash chains (stricter embedding-based groups in Sec. E.6).

Ovarian tumour ultrasound MMOTU OTU_2d [Zhao et al., 2022] (CC BY 4.0): 1,202 images with tumour masks, labelled benign cystic (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with solid components (teratoma, theca-cell tumour, mucinous cystadenoma, high-grade serous carcinoma). The thyroid detector is applied unchanged: Trap A 341/276, Trap B 91/103, caliper-free 170/178 (benign-cystic/solid).

Capsule endoscopy SEE-AI frames [Yokote et al., 2023] (CC BY 4.0) containing only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Contamination (debris, bubbles, bile) is segmented by a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62; expert masks are used where available). Debris-free: $< 3\%$ of the frame; Trap A: debris $\geq 10\%$ with $r \geq 0.5$; Trap B: $r < 0.1$. Leakage groups: blocks of 100 consecutive frame numbers plus pHash duplicates.

Chest radiography NIH ChestX-ray14 [Wang et al., 2017]; lung ROIs from the TorchXRyVision PSPNet [Cohen et al., 2022]. *Controlled cohort*: 1,906 pneumothorax images without a chest drain (NEATX annotations) and 5,718 *No Finding* images from other patients; a 100×12 px radio-opaque tube is placed at controlled overlap with the lungs (7,610 images in common support). *Real drains* (in-ROI only): 29,687 images; drains are labelled by NEATX for pneumothorax-positive images and by a drain detector (precision 0.95) for negatives; environments are built within view position (AP/PA); five patient-grouped folds.

B Why location matters: a linear-Gaussian account

Let a frozen encoder yield a disease block with class-separation signal-to-noise ratio S and an artifact block with SNR A per unit of artifact presence; training uses $P(A=1 | Y=1) = p_1$, $P(A=1 | Y=0) = p_0$ and the reversed test swaps them. For the LDA-optimal linear head the artifact’s effective training SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}}(S, \tilde{A}) = \Phi \left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}} \right), \quad (3)$$

increasing in S and decreasing in $\tilde{A}$ (verified by simulation to within 0.009 AUROC; Fig. 5). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$. Out of the ROI, $\tilde{A}_m = 0$; inside, $\tilde{A}_m \geq \tilde{A}$. Hence (i) in-ROI masking harms exactly when the mask removes disease-informative context ($S_m < S$) and helps

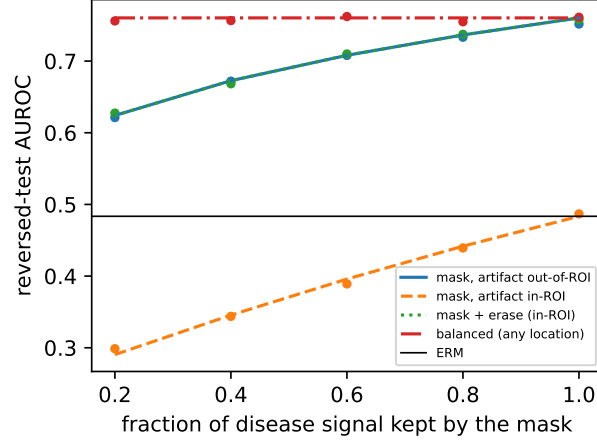


Figure 5: Reversed-test AUROC predicted by the linear-Gaussian model (lines) and simulated with logistic heads (points) as masking removes a growing share of disease signal ($S = 1$, $A = 2$, 90/10 training). Generated by `scripts/analysis/theory_sim.py`.

only by denoising ($S_m > S$); (ii) for any S_m the benefit of masking is larger out of the ROI than inside it—the crossover is always positive; (iii) erasing a subspace that contains the artifact direction sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$, where ρ is its overlap with the disease direction, so erasure beats in-ROI masking unless the artifact resembles the pathology, which disease protection prevents; (iv) group balancing removes the artifact–label association and everything correlated with it, giving $\Phi(\sqrt{S}/2)$ irrespective of location; (v) a shortcut carried by correlates that neither the mask nor pixel-level erasure touch can only be removed by label-based reweighting. Each consequence is borne out in Sections C–F.

Does the account match real encoders? For any test composition $q_y = P(A=1 \mid Y=y)$ the same head gives $\text{AUROC} = \Phi(\Delta/\sqrt{\sigma_1^2 + \sigma_0^2})$ with $\Delta = S + (p_1 - p_0)(q_1 - q_0)A/(1 + vA)$, $\sigma_y^2 = S + w_a^2(1 + Aq_y(1 - q_y))$, $w_a = \sqrt{A}(p_1 - p_0)/(1 + vA)$ and $v = [p_1(1 - p_1) + p_0(1 - p_0)]/2$. In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC only and compared the implied reversed-test AUROC with the held-out observation; no parameter is shared across models or tuned (Fig. 2). Over 148 ERM and masking models (real traps, chest drains, controlled sweeps) the mean absolute error is 0.039 AUROC ($r = 0.92$), against 0.234 for “reversed = clean” and 0.108 for a linear-symmetry heuristic (2·clean – correlated). The implied location crossover agrees with the observed one across 14 cohort–backbone pairs ($r = 0.997$, mean absolute error 0.014, 13/14 signs; the miss is a chest-radiograph cell with an observed crossover of -0.002), the implied mask gains agree with the controlled sweeps ($r = 0.965$, 42/45 signs), and the agreement extends to end-to-end fine-tuned networks (4/4 crossover signs), which violate the model’s assumptions. Because our pre-registered criterion required every crossover sign to agree, we still call the model a qualitative account. Its one clear failure is the chest-drain cohort (implied 0.70–0.76, observed 0.19–0.24): the clean and correlated environments do not expose a shortcut carried by correlates of the drain, regime (v).

C Results: dermoscopy

C.1 Masking reverses with location (controlled)

With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018 images, DINOv2 ViT-B/14 at 518px, three seeds), ROI masking improves reversed-environment AUROC over ERM by $+0.184$ [$+0.146, +0.224$] when the ruler lies outside the lesion and *lowers* it by -0.097 [$-0.138, -0.056$], -0.112 [$-0.155, -0.068$] and -0.129 [$-0.170, -0.087$] at 50, 75 and 100% overlap (location interaction $+0.313$ [$+0.272, +0.353$]; Table 5). The sign flip is

Table 5: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	rev@0	rev@50	rev@100
ERM		0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.476	0.825	0.386	+0.184 [+0.146, +0.224]	-0.097 [-0.138, -0.056]	-0.129 [-0.170, -0.087]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.759	+0.141 [+0.120, +0.162]	+0.211 [+0.187, +0.234]	+0.243 [+0.218, +0.269]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.792	0.806	0.796	+0.137 [+0.109, +0.168]	+0.219 [+0.188, +0.252]	+0.281 [+0.248, +0.315]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.091, +0.160]	+0.221 [+0.184, +0.261]	+0.279 [+0.240, +0.320]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.852	0.810	0.810	0.809	0.809	+0.167 [+0.139, +0.197]	+0.257 [+0.212, +0.263]	+0.294 [+0.265, +0.325]
Repair consistency	artifact pixels (train)	0.828	0.707	0.822	0.616	0.814	0.558	+0.042 [+0.015, +0.069]	+0.043 [+0.014, +0.072]	+0.042 [+0.012, +0.073]
Insert-then-erase (artifact templates)	artifact templates	0.848	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.152, +0.206]	+0.260 [+0.229, +0.291]	+0.303 [+0.271, +0.336]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.841	0.842	0.843	0.840	0.844	0.842	+0.177 [+0.151, +0.205]	+0.267 [+0.238, +0.298]	+0.327 [+0.293, +0.362]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.750	+0.134 [+0.115, +0.154]	+0.197 [+0.174, +0.220]	+0.215 [+0.190, +0.240]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.654	0.810	0.603	+0.061 [+0.049, +0.074]	+0.081 [+0.068, +0.094]	+0.088 [+0.077, +0.099]

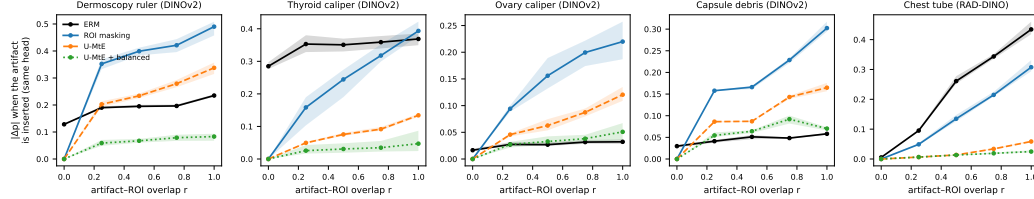


Figure 6: Mechanism: same-head counterfactual sensitivity $|p(x+\text{artifact}) - p(x)|$ versus the artifact's overlap with the ROI in the five controlled sweeps (mean over seeds; band: range over seeds). Masking removes the out-of-ROI artifact ($r = 0$) but increasingly relies on an in-ROI one; erasure and balancing remove that reliance. Generated by `scripts/analysis/mechanism_figure.py`.

reproduced at 224 px (+0.361 at 0%; -0.095 at 100%) and under randomised ruler colour, opacity, ticks and scale. At 0% overlap the masked model's correlated and reversed AUROC coincide (0.849/0.849): the shortcut is removed. At 100% the masked model's correlated AUROC exceeds ERM's (0.961 vs 0.937): it uses the ruler *more*.

Insert-then-erase in this table is the artifact-specific precursor of U-MtE: its insertion pairs use templates of the artifact type (rulers) and it erases from unmasked features. U-MtE (Sec. 4) replaces the templates by one generic overlay library for every modality and erases after masking; its dermoscopy results are in Sec. F.

C.2 Mechanism: retention, not occlusion or context

Three explanations were tested separately. *Occlusion*: with the ruler present but uncorrelated with the label (50/50), masking helps at every overlap (+0.036, +0.040, +0.039 at 0/50/100%); covering lesion pixels costs nothing measurable, so the harm requires the correlation. *Lost context*: dilating the mask at 50% overlap *increases* harm (-0.097 → -0.138 → -0.154 for margins 0/10/25 px) as the margin restores ruler pixels (retention 0.50 → 0.73 → 0.95); at 100% overlap, where retention is already 1, harm is flat (-0.129 to -0.144) although the ruler's share of visible pixels falls threefold. *Amplified reliance*: masking doubles the same-head counterfactual sensitivity to the ruler ($|\Delta p|$ 0.235 → 0.491 at 100%). Harm is largest for small lesions (-0.364 vs +0.064 for large lesions at 100%), where the ruler is a larger share of what survives the mask. Figure 6 shows the same retention mechanism in all five controlled sweeps: with the head held fixed, inserting the artifact changes the masked model's prediction more the deeper the artifact lies in the ROI. At full overlap this sensitivity exceeds that of ERM for dermoscopic rulers (0.491 vs 0.235), ovarian calipers (0.218 vs 0.032) and capsule debris (0.302 vs 0.058) and matches it for thyroid calipers (0.393 vs 0.369). In chest radiography ERM ignores a tube outside the lungs ($|\Delta p| = 0.005$ at $r = 0$), which is why masking has nothing to remove there. U-MtE lowers the masked model's sensitivity in every sweep (full overlap: 0.338, 0.134, 0.120, 0.164, 0.059), and U-MtE with group balancing lowers it most (≤ 0.083).

C.3 Real hair (ISIC 2019), pre-registered

Two traps share one hair-free group (at most 30 hair pixels): Trap A contains hair lying mostly inside the lesion ($r \geq 0.5$), Trap B hair mostly outside ($r < 0.1$); cells are matched to the hair-free source mix, and five seeds of five-fold group-safe cross-validation give 172 reversed-test melanomas per trap. Masking helps in Trap B (+0.107 [+0.078, +0.135]), harms in Trap A (-0.037

402 $[-0.057, -0.019]$, negative in all five seeds) and the crossover is $+0.145 [+0.105, +0.181]$. The
403 harm holds in BCN images ($-0.051 [-0.075, -0.027]$); in HAM it is not distinguishable from
404 zero ($-0.009 [-0.036, +0.017]$). Oracle hair inpainting helps in both traps ($+0.103, +0.096$).
405 Group-balanced training and DFR, which need only an image-level hair label, give the largest gains
406 ($+0.225/+0.251$ in Trap A, $+0.209/+0.215$ in Trap B). Paired linear erasure recovers less than
407 half of that ($+0.086/+0.081$). The magnitude of the in-ROI harm depends on the definition of the
408 comparison group: with a looser hair-free group ($< 0.1\%$ of pixels) and only heavily haired images,
409 masking gives no benefit but no measurable harm inside the lesion ($+0.012 [-0.042, +0.062]$) while
410 still helping outside ($+0.178$); the benefit shrinks to zero as more of the hair lies inside the lesion.

411 With the dermatology foundation model DermLIP, the controlled ruler sweep reverses as with DI-
412 NOv2 ($+0.161$ at no overlap, -0.263 to -0.309 at 25–100%), but on real hair masking harms
413 at both locations (Trap A $-0.085 [-0.107, -0.063]$, Trap B -0.088): with this encoder masking
414 removes disease context regardless of where the hair lies, so no location gap appears.

415 C.4 What the real-hair shortcut is made of

416 Removing hair pixels at test time only closes 44–47% of the ERM clean–reversed gap: most of
417 what a frozen encoder exploits in a real hair trap is not the hair pixels but what accompanies hair
418 (hair-bearing images differ in sex, age and body site). This bounds every pixel-level method, in-
419 cluding oracle inpainting and U-MtE, and explains why image-level-label methods, which remove
420 everything correlated with the label, recover more.

421 C.5 Reversed-environment gains can be gamed

422 Erasing the artifact concept from unpaired image-level labels yields the largest reversed gain in the
423 study ($+0.326$ in Trap A) while *inverting* the correlated/reversed ordering (0.618 vs 0.817). Post-
424 hoc prevalence-equalised recalibration [Kina and Petersen, 2026] behaves the same way under the
425 trap’s 90/10 shift ($+0.427$; correlated 0.433 vs reversed 0.918). Both remove diagnosis-correlated
426 signal or over-correct the group prior rather than removing reliance on the artifact; we therefore read
427 every reversed gain together with clean AUROC and the correlated/reversed ordering.

428 D Results: ultrasound, capsule endoscopy and chest radiography

429 D.1 Thyroid ultrasound: sonographer calipers

430 TN3K images with TNCD labels (3,493 images, 1,210 malignant) carry measurement calipers that
431 a sonographer places on or around the nodule. Calipers were segmented by an audited rule-based
432 detector (image-level precision and recall 0.86 – 0.89). With the E13 protocol unchanged, masking
433 the nodule helps greatly when the calipers lie outside it (Trap B: $+0.341 [+0.320, +0.363]$) and
434 much less when they lie inside it (Trap A: $+0.111 [+0.097, +0.124]$); the crossover is $+0.230$
435 $[+0.204, +0.255]$ with DINOv2 and $+0.163 [+0.132, +0.194]$ with MedSigLIP. After masking,
436 the Trap A model still relies on the calipers (correlated 0.902 versus reversed 0.400). With syn-
437 thetic calipers swept across the nodule boundary (1,663 caliper-free images), masking gains $+0.402$
438 $[+0.347, +0.458]$ at no overlap and $-0.030 [-0.079, +0.016]$ at full overlap (interaction $+0.432$
439 $[+0.377, +0.492]$): the entire benefit of masking disappears, without significant harm.

440 D.2 Capsule endoscopy: intestinal debris

441 SEE-AI frames (erosion 3,550 versus polyp-like 1,931; expert lesion boxes as ROI) with contamina-
442 tion masks from a patch-token probe trained on 2,160 expert-masked frames (held-out pixel accuracy
443 0.92 , IoU 0.62). Leakage groups are contiguous blocks of 100 frame numbers plus perceptual-hash
444 duplicates (no video identifiers are published). Masking helps against out-of-ROI debris ($+0.465$
445 $[+0.440, +0.490]$) and much less against debris on the lesion ($+0.097$), a crossover of $+0.368$
446 $[+0.340, +0.397]$. With synthetic debris on the 796 cleanest frames the reversal appears: $+0.103$
447 $[+0.060, +0.149]$ at no overlap, $-0.102 [-0.162, -0.040]$ at full overlap (interaction $+0.205$
448 $[+0.150, +0.259]$).

D.3 Chest radiography: a boundary case

With a synthetic tube swept into the lung fields of NIH pneumothorax radiographs (RAD-DINO), the encoder does not use an extra-pulmonary tube at all (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove; inside the lungs the tube is used (reversed 0.614) and lung masking only partly removes it (0.725). On real devices (RANZCR-CLiP linked to NIH; central lines and endotracheal tubes) masking helped for both locations. The location-dependent *harm* of masking is therefore not universal; what is universal in our data is that masking leaves an in-ROI shortcut largely intact.

E Results: natural distribution, fine-tuning and label robustness

E.1 Unaltered test distributions

Without any resampling, in-ROI artifacts are naturally associated with the label in every cohort (thyroid calipers: 37.6% of benign vs 24.1% of malignant nodules; in-lesion hair: 31% of melanomas vs 20% of benign lesions; capsule debris on the lesion: 23% of polyp-like lesions vs 10% of erosions). We trained on natural training data and tested on untouched data: the official TN3K test split, each ISIC source held out in turn (a real change of hospital and device), and held-out capsule frame blocks. On the shortcut-conflicting pairs, masking *lowers* AUROC for thyroid (−0.102 [−0.139, −0.066]) and for two of the three held-out hospitals (BCN −0.016 [−0.027, −0.005], MSK −0.056 [−0.084, −0.029]); it helps for HAM (+0.053) and does not differ significantly from ERM on capsule. U-MtE with group balancing improves on masking for these pairs in every cohort (+0.014 to +0.088) and gives the best or tied-best cross-group AUROC in three of five settings (HAM 0.786, BCN 0.734 vs 0.735, capsule 0.947); plain group balancing is best on thyroid and MSK. Without artifact labels, U-MtE improves on masking for thyroid (+0.051) and HAM (+0.023) only.

Clinical secondary metrics Table 6 reports AUPRC, Brier score, expected calibration error and sensitivity/specificity at the operating point fixed on validation data. On the official thyroid split masking lowers sensitivity from 0.734 to 0.347 while specificity rises from 0.619 to 0.888 (balanced accuracy 0.677 → 0.618); U-MtE recovers part of the sensitivity (0.425) and has the best AUPRC (0.638) and Brier score (0.203). For the held-out BCN hospital masking worsens both Brier score (0.239 → 0.286) and calibration (ECE 0.279 → 0.332); for HAM, MSK and capsule it improves every metric, in line with the AUROC results.

E.2 End-to-end fine-tuning

Fine-tuning ResNet-50 end-to-end (five folds) reproduces the frozen-backbone findings. Capsule: masking harms in-ROI (−0.284 [−0.347, −0.226]) and helps out-of-ROI (+0.301, crossover +0.584 [+0.469, +0.691]). Thyroid: the crossover points the same way (+0.057 [−0.021, +0.130], five folds), and an end-to-end analogue of U-MtE—an invariance penalty between masked images with and without a generic overlay—improves on masking by +0.143 [+0.099, +0.188] without shortcut flipping (reversed 0.679, correlated 0.703), matching the best label-using arm. It fails on capsule (−0.021), where the artifact resembles the pathology. Ovary (1,202 images) is underpowered: U-MtE end-to-end has the best reversed, correlated and clean AUROC, with CIs overlapping masking. A fine-tuned ViT-S reproduces the crossover on thyroid (+0.168 [+0.074, +0.265]), but the end-to-end U-MtE analogue does not improve on masking with this architecture (+0.009 [−0.021, +0.040]). Applying U-MtE post hoc to the penultimate features of the masked, fine-tuned ResNet-50 does not help either (pre-registered; +0.004 [+0.000, +0.012] over the same features without erasure): after task training the shortcut is encoded along task-specific directions that generic overlays do not move. For fully fine-tuned networks the subspace must be estimated and removed *during* training. *Erase-while-fine-tuning* (pre-registered; no artifact annotation) projects a running estimate of the overlay-shift subspace of the network’s own penultimate features out before the head at every step and at test time, and adds feature invariance and prediction consistency between masked images with and without a generic overlay. On thyroid it improves on masking by +0.157 [+0.114, +0.205] with ResNet-50 and +0.143 [+0.103, +0.187] with ViT-S (reversed AUROC 0.693 and 0.608 vs 0.536 and 0.465), whereas each component alone works for only one architecture (projection + invariance: ResNet-50 +0.160 [+0.114, +0.207], ViT-S +0.061 [−0.005, +0.141];

Table 6: Clinical secondary metrics on the unaltered test sets (Sec. E): mean (SD) over training seeds. Sensitivity and specificity at the operating point fixed on clean validation data; ECE with 10 equal-width bins; AUPRC baseline = prevalence.

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.736 (0.009)	0.606 (0.012)	0.207 (0.005)	0.082 (0.008)	0.734 (0.047)	0.619 (0.061)
	mask	0.731 (0.006)	0.632 (0.007)	0.214 (0.006)	0.120 (0.026)	0.347 (0.095)	0.888 (0.041)
	U-MtE	0.742 (0.003)	0.638 (0.007)	0.203 (0.001)	0.085 (0.007)	0.425 (0.088)	0.854 (0.050)
	balanced	0.716 (0.007)	0.584 (0.012)	0.219 (0.004)	0.110 (0.007)	0.714 (0.052)	0.599 (0.061)
	U-MtE _{bal}	0.731 (0.004)	0.630 (0.007)	0.207 (0.001)	0.084 (0.006)	0.399 (0.069)	0.866 (0.034)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.586 (0.003)	0.239 (0.006)	0.279 (0.011)	0.842 (0.038)	0.519 (0.075)
	mask	0.769 (0.004)	0.571 (0.009)	0.286 (0.007)	0.332 (0.009)	0.860 (0.013)	0.439 (0.023)
	U-MtE	0.773 (0.004)	0.574 (0.008)	0.280 (0.009)	0.328 (0.012)	0.858 (0.039)	0.448 (0.089)
	balanced	0.781 (0.005)	0.585 (0.009)	0.228 (0.006)	0.257 (0.011)	0.811 (0.046)	0.554 (0.087)
	U-MtE _{bal}	0.774 (0.004)	0.568 (0.008)	0.279 (0.011)	0.325 (0.016)	0.860 (0.027)	0.447 (0.051)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.172 (0.007)	0.161 (0.019)	0.608 (0.049)	0.786 (0.039)
	mask	0.790 (0.004)	0.512 (0.009)	0.146 (0.003)	0.115 (0.009)	0.593 (0.052)	0.824 (0.037)
	U-MtE	0.783 (0.004)	0.500 (0.008)	0.150 (0.003)	0.126 (0.009)	0.566 (0.073)	0.831 (0.049)
	balanced	0.766 (0.008)	0.462 (0.014)	0.178 (0.008)	0.170 (0.020)	0.626 (0.047)	0.755 (0.046)
	U-MtE _{bal}	0.774 (0.006)	0.482 (0.009)	0.154 (0.004)	0.131 (0.010)	0.594 (0.061)	0.798 (0.051)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.750 (0.005)	0.353 (0.005)	0.162 (0.006)	0.234 (0.011)	0.535 (0.092)	0.813 (0.068)
	mask	0.789 (0.002)	0.404 (0.006)	0.143 (0.005)	0.217 (0.011)	0.559 (0.112)	0.842 (0.063)
	U-MtE	0.799 (0.008)	0.406 (0.009)	0.145 (0.007)	0.226 (0.016)	0.573 (0.067)	0.846 (0.041)
	balanced	0.747 (0.017)	0.346 (0.019)	0.156 (0.014)	0.228 (0.015)	0.539 (0.078)	0.803 (0.076)
	U-MtE _{bal}	0.802 (0.005)	0.401 (0.009)	0.139 (0.006)	0.215 (0.021)	0.606 (0.054)	0.825 (0.037)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.859 (0.032)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.871 (0.042)	0.897 (0.043)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.758 (0.087)	0.838 (0.059)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.044 (0.012)	0.866 (0.048)	0.897 (0.052)

501 prediction consistency: ResNet-50 +0.007 [−0.014, +0.029], ViT-S +0.062 [+0.031, +0.087]).
502 With ResNet-50 the correlated AUROC falls slightly below the reversed one (0.661 vs 0.693), a
503 mild over-correction.

504 E.3 Zero-shot vision–language models

505 Without any task training, the foundation models’ own zero-shot scores (similarity to “benign” ver-
506 sus “malignant” text prompts) are already swayed by artifacts: on capsule frames the correlated and
507 reversed AUROC differ by 0.40. Masking changes these zero-shot scores exactly as the location law
508 predicts: for MedSigLIP it leaves reversed AUROC unchanged for in-ROI artifacts but improves it
509 for out-of-ROI artifacts (thyroid +0.013 vs +0.116, crossover +0.103 [+0.079, +0.127]; capsule
510 crossover +0.049 [+0.024, +0.076]), and it harms more for in-ROI artifacts on ovarian ultrasound
511 (crossover +0.181 [+0.135, +0.228]) and for DermLIP on hair (+0.051 [+0.031, +0.071]). The
512 location effect is therefore a property of the representation, not of the classifier trained on top of it.

513 E.4 Robustness to artifact-label definitions

514 Artifact masks for calipers come from a detector (visual audit: 95% of thyroid detections lie on
515 real calipers; 92–97% of caliper-free images contain no visible caliper). Re-running the thyroid and
516 ovary analyses with large-marker-only (≥ 50 px) and strict-location ($r \geq 0.7 / r < 0.05$) definitions
517 leaves the crossover (thyroid +0.215 to +0.262; ovary +0.170 to +0.195) and the U-MtE_{protect} gain
518 over masking (thyroid +0.217 to +0.282; ovary +0.043 to +0.067) significant in every variant; both
519 grow under stricter definitions, as expected if detector errors dilute rather than create the effects.

520 E.5 Backbone scale

521 Replacing DINOv2 ViT-B/14 (86 M parameters) by ViT-S/14 (21 M) or ViT-L/14 (304 M), with ev-
522 erything else fixed (pre-registered), the location crossover is positive in all nine cohort–size combina-
523 tions (ViT-S / ViT-B / ViT-L: thyroid +0.244 / +0.230 / +0.271; ovary +0.298 / +0.170 / +0.202;
524 capsule +0.370 / +0.368 / +0.408; every CI excludes zero), and protected U-MtE beats masking
525 on thyroid and ovary at every size (ViT-S +0.181 [+0.146, +0.218] and +0.103 [+0.078, +0.128];
526 ViT-L +0.222 [+0.195, +0.256] and +0.060 [+0.043, +0.077]). Larger models exploit the in-ROI

Table 7: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by scripts/make_main_table.py.

Method	Needs	Hair/DINOv2	Hair/DeiT-LIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConcNet	Ovary/DINOv2	Ovary/MedSigLIP	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConcNet	Derm/RAD-DINO	Derm/DINOv2
ERM		0.49 (0.49)	0.67 (0.67)	0.29 (0.29)	0.28 (0.28)	0.35 (0.35)	0.41 (0.41)	0.42 (0.42)	0.59 (0.59)	0.59 (0.59)	0.55 (0.55)	0.19 (0.19)	0.28 (0.28)
ROI masking	ROI	0.45 (0.45)	0.58 (0.58)	0.40 (0.40)	0.35 (0.35)	0.36 (0.36)	0.55 (0.55)	0.43 (0.43)	0.68 (0.68)	0.70 (0.70)	0.64 (0.64)	0.24 (0.24)	0.29 (0.29)
Mask + overlay augmentation	ROI	0.46 (0.46)		0.42 (0.42)	0.38 (0.38)	0.37 (0.37)	0.53 (0.53)		0.70 (0.70)	0.72 (0.72)	0.65 (0.65)	0.24 (0.24)	0.29 (0.29)
U-MtE (ours)	ROI	0.51 (0.51)	0.51 (0.51)	0.62 (0.62)	0.55 (0.55)	0.49 (0.49)	0.56 (0.56)	0.64 (0.64)	0.57 (0.57)	0.47 (0.47)	0.56 (0.56)	0.27 (0.27)	0.28 (0.28)
U-MtE, protected (ours)	ROI + Y of A=0	0.49 (0.49)	0.59 (0.59)	0.62 (0.62)	0.61 (0.61)	0.55 (0.55)	0.59 (0.59)	0.57 (0.57)	0.67 (0.67)	0.67 (0.67)	0.64 (0.64)	0.27 (0.27)	0.28 (0.28)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.66 (0.66)	0.71 (0.71)	0.73 (0.73)	0.78 (0.78)	0.81 (0.81)	0.72 (0.72)	0.54 (0.54)	0.40 (0.40)
DFR	A	0.74 (0.74)	0.78 (0.78)	0.74 (0.74)	0.76 (0.72)	0.71 (0.67)	0.72 (0.69)		0.80 (0.80)	0.80 (0.80)	0.77 (0.77)	0.68 (0.68)	0.55 (0.55)
Mask + DFR	ROI + A	0.69 (0.69)		0.83 (0.74)	0.81 (0.76)	0.77 (0.72)	0.75 (0.75)	0.88 (0.88)	0.90 (0.90)	0.86 (0.86)	0.65 (0.65)	0.57 (0.57)	
U-MtE + balanced (ours)	ROI + A	0.70 (0.70)	0.65 (0.65)	0.79 (0.79)	0.79 (0.79)	0.74 (0.74)	0.74 (0.74)	0.79 (0.79)	0.86 (0.86)	0.85 (0.85)	0.80 (0.80)	0.46 (0.46)	0.44 (0.44)

shortcut *more*: the ERM correlated–reversed gap grows from 0.547 to 0.621 (thyroid) and from 0.319 to 0.570 (ovary) between ViT-S and ViT-L, so scale does not remove it. On capsule, disease protection is less complete with ViT-L (−0.040 [−0.046, −0.035] versus masking).

E.6 Stricter leakage grouping

TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). As a pre-registered sensitivity analysis we united these groups with connected components of mean-centred DINOv2 embeddings at the most aggressive cosine threshold that keeps every group below 5% of the cohort (thyroid: 3,455 → 3,081 groups, $\tau = 0.80$; ovary: 1,059 → 969, $\tau = 0.80$; capsule: 137 → 135, $\tau = 0.93$) and re-ran all primary contrasts. Eleven of twelve keep their sign and significance: the location crossover is +0.235 [+0.211, +0.259] (thyroid), +0.157 [+0.080, +0.235] (ovary) and +0.394 [+0.359, +0.430] (capsule), and protected U-MtE still beats masking on thyroid (+0.251 [+0.219, +0.279]) and ovary (+0.047 [+0.020, +0.072]). The exception is the small ovary advantage of erasure over augmentation, which loses significance (+0.020 [−0.011, +0.053]). Embedding grouping removes visually near-identical images of one patient or session; it cannot exclude differently looking images of the same patient.

F Results: artifact-agnostic mask-then-erase

F.1 U-MtE repairs what masking leaves behind

Without any artifact example, artifact mask or artifact label—it uses only the ROI mask that masking already requires—U-MtE improves reversed AUROC over masking in the in-ROI regime of every modality: thyroid calipers +0.222 [+0.197, +0.251] (DINOv2) and +0.204 [+0.174, +0.237] (MedSigLIP); dermoscopic hair +0.052 [+0.044, +0.062]; synthetic capsule debris +0.081 [+0.050, +0.112]; synthetic calipers +0.278 [+0.232, +0.325]; synthetic chest tubes +0.134 [+0.120, +0.151] (DINOv2 / RAD-DINO). With MedSigLIP the unbalanced method *loses* to masking in both controlled sweeps (debris −0.114 [−0.147, −0.082], calipers −0.076 [−0.110, −0.043]) although it wins on real thyroid calipers (+0.204); the balanced variant beats masking in all four controlled sweeps (+0.193 to +0.471) and has no location interaction in any of them. Disease protection turns the MedSigLIP debris failure into a gain (+0.061 [+0.032, +0.092] over masking) but not the caliper one (−0.041 [−0.089, −0.005]).

Sensitivity to the erasure rank In a pre-registered ablation (Trap A, DINOv2), the thyroid gain over masking is +0.18 to +0.22 for every setting that erases at least 16 directions (held-out energy 0.5–0.99, or fixed $k \in \{16, 64\}$), but only +0.06 to +0.07 with one to four directions: the shift caused by overlays is multi-dimensional. The pre-specified 90% rule reached the rank cap ($k = 64$) in all three cohorts. The protected variant never harms capsule by more than 0.010 and helps ovary at every rank (+0.007 to +0.043), whereas unprotected erasure harms capsule more the more directions it removes, as the overlap term ρ of Sec. B predicts.

563 F.2 Erasure, not augmentation

564 Using the same generic overlays as training augmentation (overlaid copies of a random half of
565 the training images, in the masked view) gains at most +0.03 over masking; erasing their sub-
566 space instead gains +0.207 [+0.182, +0.235] (thyroid, DINOv2), +0.177 [+0.148, +0.211] (thy-
567 roid, MedSigLIP) and +0.045 [+0.037, +0.054] (hair) more than augmentation. Under the stricter
568 near-duplicate grouping (Sec. E.6) the thyroid advantage is unchanged (+0.192) but the small ovary
569 advantage (+0.035) loses significance, so we regard erase > augment as established for thyroid and
570 hair only.

571 F.3 When the artifact resembles the disease

572 On real capsule debris U-MtE is worse than masking (−0.115): erosions carry yellow-white
573 fibrin that resembles debris, and a subspace estimated from real-debris templates is worse still
574 (−0.395). Disease protection removes the failure (−0.010 [−0.020, −0.001] generic, −0.012
575 [−0.028, +0.003] template) at a cost of at most 0.012 where erasure already works (thyroid −0.005
576 [−0.036, +0.024], hair −0.012 [−0.020, −0.003]).

577 F.4 With image-level artifact labels

578 Adding group balancing, U-MtE_{bal} is the most robust model in thyroid ultrasound by the minimum
579 of correlated and reversed AUROC (0.788 DINOv2, 0.792 MedSigLIP; DFR 0.659 and 0.719); in
580 real capsule endoscopy masking plus last-layer retraining is higher (0.885 versus 0.857); on der-
581 moscopic hair DFR alone is best (0.741) because most of the hair shortcut is not in the hair pixels
582 (removing every hair patch with the oracle mask gains only +0.015). In both controlled sweeps the
583 benefit of U-MtE_{bal} no longer depends on location (interaction −0.000 [−0.023, +0.023] capsule;
584 −0.054 [−0.098, −0.004] thyroid), whereas that of masking does (+0.205, +0.432). The capsule
585 findings replicate with MedSigLIP (crossover +0.332 [+0.296, +0.365]; masking plus DFR 0.901).
586 All thyroid and capsule findings replicate with a CNN (ConvNeXt-Base): crossovers +0.284 and
587 +0.292; U-MtE − mask +0.128 [+0.112, +0.145] and erase − augment +0.120 [+0.102, +0.139]
588 on thyroid.

589 F.5 Label-free and erasure baselines

590 Among methods that need no artifact labels, U-MtE beats JTT [Liu et al., 2021] by +0.198
591 [+0.167, +0.231] (thyroid), +0.137 [+0.098, +0.179] (ovary) and +0.094 [+0.077, +0.109]
592 (chest drains, RAD-DINO; not with DINOv2, −0.010); JTT on the unmasked view is best on hair.
593 Task-preserving concept removal (SPLINCE; Holstege et al., 2025), fitted with artifact labels in
594 the whitened, minimal-distortion form of the original method, removes the artifact covariance at a
595 large cost in disease signal (clean AUROC 0.56–0.78 versus 0.69–0.87 for masking) and still flips
596 the shortcut on capsule endoscopy (reversed 0.868, correlated 0.683): when artifact and label are
597 strongly associated in training, removing one removes much of the other. By min(reversed, corre-
598 lated) U-MtE_{bal} exceeds SPLINCE on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683;
599 0.704 vs 0.564); on chest drains SPLINCE is higher (0.609 vs 0.460) but below DFR (0.683).

600 F.6 Text-prompted erasure

601 With vision–language backbones an artifact direction can be obtained from text alone (“a thyroid
602 ultrasound image with measurement calipers” minus “a thyroid ultrasound image”; biased-prompt
603 construction, Chuang et al., 2023). Erasing that subspace after masking is a near no-op in every
604 cohort (+0.001 to +0.008 over masking; MedSigLIP thyroid, ovary, capsule and DermLIP hair),
605 whereas U-MtE with the same MedSigLIP features gains +0.204 [+0.174, +0.237] (thyroid) and
606 +0.215 [+0.182, +0.246] (ovary). The direction along which a described artifact moves the *text*
607 embedding is not the direction along which the real artifact moves the *image* embedding; synthetic
608 insertions measure the latter directly.

F.7 A single procedure: validation-driven selection

Selecting, per fold, the candidate with the best group-aware validation score (image-level artifact labels needed on validation images only) beats masking in all five cohorts (+0.17 to +0.40), is within 0.02 of the best candidate in each, and beats DFR in three (thyroid +0.057, capsule +0.047, ovary +0.010); it chooses U-MtE_{bal} where the artifact is an overlay and plain balancing on hair and drains, where DFR remains better. A split-validation variant that admits DFR as a candidate is reported in the supplement.

F.8 Modern inpainting

Inpainting the detected caliper pixels with LaMa [Suvorov et al., 2022] and then masking is a strong baseline when artifact masks are available (thyroid +0.226 [+0.213, +0.238], ovary +0.123 [+0.089, +0.155] over masking). U-MtE_{protect}, which needs no artifact mask, matches it on thyroid (−0.009 [−0.045, +0.023]) and is below it on ovary (−0.080) and on dermoscopic hair, where published hair masks are available (−0.086); with image-level artifact labels, U-MtE_{bal} exceeds it in all three (+0.162, +0.067, +0.123).

G Statistical plan and pre-registration

Every experiment was specified in a dated, publicly versioned document before any model was fitted on its data. The 16 primary tests (location crossover, protected U-MtE vs masking, erase vs augment, U-MtE_{bal} vs balanced; four cohorts each) are Holm-corrected; 12/16 are supported (Fig. 4). Everything else is secondary. Datasets considered and not used (CANDID-PTX: no access; breast ultrasound: no artifact-free group; colonoscopy: no reflection-free images; ISIC ink, cardiomegaly lines: too few images; pathology pen marks: unreliable masks) and the leakage analysis are documented in the repository.

H Compute

All experiments ran on one RTX 3090 (24 GB), 8 CPU cores and 31 GB RAM; the full pipeline takes about two days, dominated by feature extraction and fine-tuning.

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